

Report of **Sickle cell symposium**

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Introduction

Sickle cell disease (SCD), also known as sickle cell anaemia, is a genetic blood disorder characterised by the presence of abnormal haemoglobin, a protein that enables red blood cells to carry oxygen throughout the body. This condition is inherited when a person receives two abnormal haemoglobin genes, one from each parent.

The chosen theme for the Sickle Cell Symposium, "The Importance of Genotype Awareness on Personal Health," underscores the critical role of understanding one's genetic makeup in the context of sickle cell disease. This theme carries several key implications for individuals within the LASU community and the broader healthcare landscape as it helped us achieve the following:

- Genotype education and awareness
- Empowering individuals to make informed decisions.
- Emphasising on ethical considerations, such as confidentiality.



Background

Globally, sickle cell disease remains a pressing health concern with the degree of prevalence varying in several regions of the world.

Here is an overview of the global prevalence and demographics of SCD as of 2022:

- It is estimated that around 300,000 infants are born with SCD globally each year.
- Sickle cell disease is most prevalent in sub-Saharan Africa, where approximately 75% of all affected births occur. Nigeria has one of the highest prevalence rates in the world.
- Other regions with a significant prevalence of SCD include the Middle East, India, and parts of the Mediterranean.

Demographics:

- Sub-Saharan Africa: SCD is most common in populations with a high frequency of the genetic mutation that causes



the disease. In many sub-Saharan African countries, a significant portion of the population carries the sickle cell trait.

- Middle East and India: Certain regions in the Middle East, as well as India, have a notable prevalence of SCD. The migration patterns of populations from these regions contribute to the presence of the disease in other parts of the world.

- Mediterranean Region: SCD is also found in populations with origins in the Mediterranean region, including parts of Southern Europe, the Middle East, and North Africa.

- United States: In the United States, SCD is most commonly found among individuals of African descent. It also affects individuals of Hispanic, Mediterranean, and Middle Eastern heritage.



- Latin America: SCD can be found in various Latin American countries, with varying prevalence rates. Brazil has a significant number of individuals affected by the disease.

It is worthy to note that sickle cell disease is inherited in an autosomal recessive pattern, meaning that an individual needs to inherit two copies of the abnormal haemoglobin gene (one from each parent) to have the disease.

The prevalence of the sickle cell trait is higher in regions where malaria is or has been endemic. Having one copy of the gene may provide some protection against malaria, leading to the persistence of the trait in these populations.

Objectives of the Symposium

- To highlight the importance of genotype awareness in managing sickle cell disease effectively.
- To raise awareness about sickle cell disease among members of the LASU community.
- To highlight the importance of genetic counselling and genotype education.

Sessions and topics covered

The symposium comprised two main sessions: a keynote session and a panel session.

Dr. Abosede Wellington delivered the keynote speech during the designated keynote session of the symposium.

The panel session, expertly moderated by Odumosu Kehinde, featured distinguished speakers, including Ms. Timileyin Ishola, a passionate sickle cell warrior and advocate; Mrs. Mujidat O. Subair, a Lawyer and Lecturer at LASU; and Ms. Omolola Olasumbo Kolade, a dedicated sickle cell warrior, advocate and entrepreneur.

Mrs. Mujidat addressed inquiries related to ethical considerations, including confidentiality. Ms. Timileyin provided insights into safeguarding the mental health of sickle cell disease (SCD) warriors, and Ms. Olasumbo shared her experiences and advice on living with SCD while excelling as an entrepreneur.

The concluding session of the event featured the founder of One Health Lifesavers, an NGO dedicated to educating individuals on the significance of blood donation. During this session, the attendees were enlightened about the



importance and benefits of contributing to blood donation efforts.

In addition, we provided complimentary routine medications, including paludrine, folic acid, and vitamin B complex, to individuals living with sickle cell disease.

However, a notable challenge we observed is the reluctance of some individuals with sickle cell disease to adhere to routine medication regimens, and, furthermore, some faced difficulties affording essential drugs such as paludrine.

Attendance, participant feedback and responses

The symposium drew a total attendance of one hundred individuals, including notable figures such as Dr. Ariyibi, the Director of the Lagos State University Health Centre, representatives from the Lagos State Ministry of Health, a representative of the Dean of the Faculty of Law, and delegates from various faculties, and the college of medicine in Ikeja.



We received positive feedback from participants, and the most impactful responses came from individuals who shared their transformation from genotype illiteracy to genotype literacy. These testimonials underscored the meaningful impact of our efforts.

Here's a link to the video made by the moderator who shared how she had become more enlightened:

https://www.instagram.com/reel/Czv516dl4yL/?utm_source=ig_web_copy_link&igshid=MzRlODBiNWFiZA==

Conclusion

In conclusion, the SCD Symposium, centred around the theme "Importance of Genotype Awareness on Personal Health," achieved its goal of enlightening the LASU community. Dr. Abosede Wellington's insightful keynote and the panel discussion, moderated by Odumosu Kehinde, provided valuable perspectives. The engagement of attendees, including Dr. Ariyibi and representatives from diverse faculties, underscored the significance of genotype awareness. Testimonials revealed a positive impact, emphasising the need for continued efforts in education, advocacy, and support for individuals navigating the complexities of sickle cell disease.